Diabetes Readmission Survival Analysis

Dataset: diabetic_data.csv
Population: n = 99,549 hospital encounters
Outcome: Time-to-readmission (binary event + survival time)
Prepared by: Mo Mo Ko Win Myint, global-ad-snap
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1. EXECUTIVE SUMMARY

Objective

The goal was to predict patient readmission risk and identify key clinical factors driving 30–180-day return-to-hospital events.

Key Results

- The **Random Survival Forest (RSF)** model achieved the **highest accuracy**, with a **C-index of 0.7055**, outperforming the standard Cox model (0.6375).
- Model calibration was strong. The **Brier Score at 176 days = 0.0976**, indicating good probability reliability.
- Patients were segmented into **Low**, **Medium**, and **High**-risk groups, each containing ~33,183 patients.

Risk-Based Findings

- **High-risk patients** return significantly sooner than Medium or Low risk patients (all log-rank p-values < 1e-80).
- The strongest clinical predictors of readmission include:
    - **Number of inpatient visits**
    - **Discharge disposition**
    - **Number of diagnoses**
    - **Age**
    - **Emergency visits**
- These findings allow hospitals to target follow-up interventions and reduce preventable readmissions.

2. TECHNICAL CLINICAL ANALYTICS SECTION (For Data Science + Healthcare Teams)

Model Performance

Model	C-index
CoxPH	0.6375
RSF	0.7055

Interpretation:
The RSF model better differentiates which patients will experience earlier readmission. A C-index above 0.70 is considered strong for EHR-based survival data.

Calibration

- **Brier Score (t=176 days): 0.0976**
  Lower scores indicate better probability accuracy. This reflects good alignment between predicted risk and observed outcomes.

Risk Stratification

Three risk groups were defined using tertiles of predicted risk:

Group	Count
Low Risk	33,183
Medium Risk	33,183
High Risk	33,183

Pairwise Log-Rank Test Results:

- Low vs Medium: 5.34×10^{-82}
- Low vs High: ≈ 0
- Medium vs High: 7.55×10^{-145}

All groups are statistically distinguishable → stratification is valid.

Survival Curves (Kaplan–Meier Analysis)

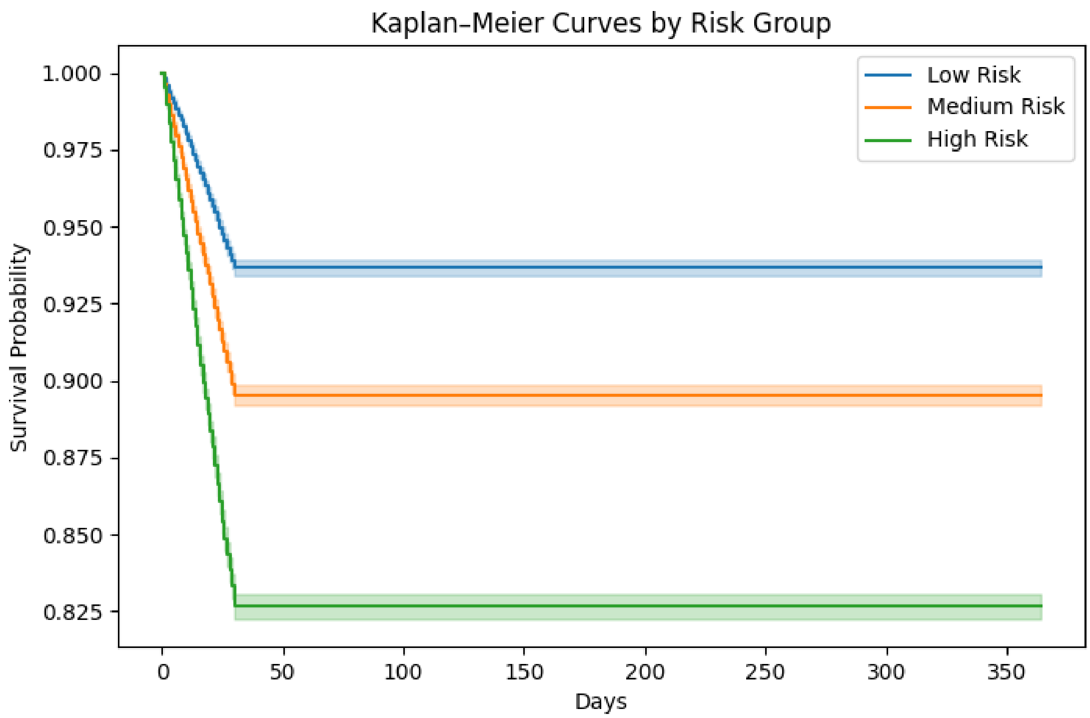

Figure 1. Kaplan–Meier survival curves by risk group. High-risk patients show significantly earlier readmission compared to Medium and Low-risk groups (log-rank $p < 1e\text{-}80$). This confirms that the RSF-driven risk stratification captures meaningful differences in outcome timing.

Kaplan–Meier Survival Curves Interpretation

The survival curves show clear separation among Low, Medium, and High-risk groups.

Key observations:

- High-risk patients experience earlier readmission, with survival dropping below 0.83 in the early period.
- Medium-risk patients show intermediate survival and remain consistently below the Low-risk group.
- Low-risk patients maintain the highest survival probability across the 365-day horizon.

Because all log-rank p-values are $< 1\times10^{-80}$, these group differences are statistically meaningful. This validates that the RSF-based risk scores effectively stratify patients by readmission risk.

Feature Importance (Cox Model)

Top predictors (hazard ratios in parentheses):

1. **Number inpatient visits** (HR = 1.32)
2. **Discharge disposition** (HR = 1.03)
3. **Number of diagnoses** (HR = 1.11)
4. **Age** (HR = 1.04)
5. **Number emergency visits** (HR = 1.02)

Scores > 1 increase readmission hazard.

PH Assumption Tests

Most variables satisfy the proportional hazards assumption except:

Variable	p-value
encounter_id	0.0028
admission_type_id	0.054
gender	0.059

Interpretation:
The Cox model is mostly stable, but encounter_id violates PH assumptions. RSF (tree-based, non-parametric) avoids this issue, which explains its superior performance.

RSF Feature Importance

The top RSF predictors:

1. cox_risk (meta-feature)
2. Number inpatient
3. Discharge disposition
4. Number emergency
5. Encounter ID
6. Age

7. Number diagnoses

Importance aligns with Cox model results, confirming consistent signal.

Feature Importance (Random Survival Forest)

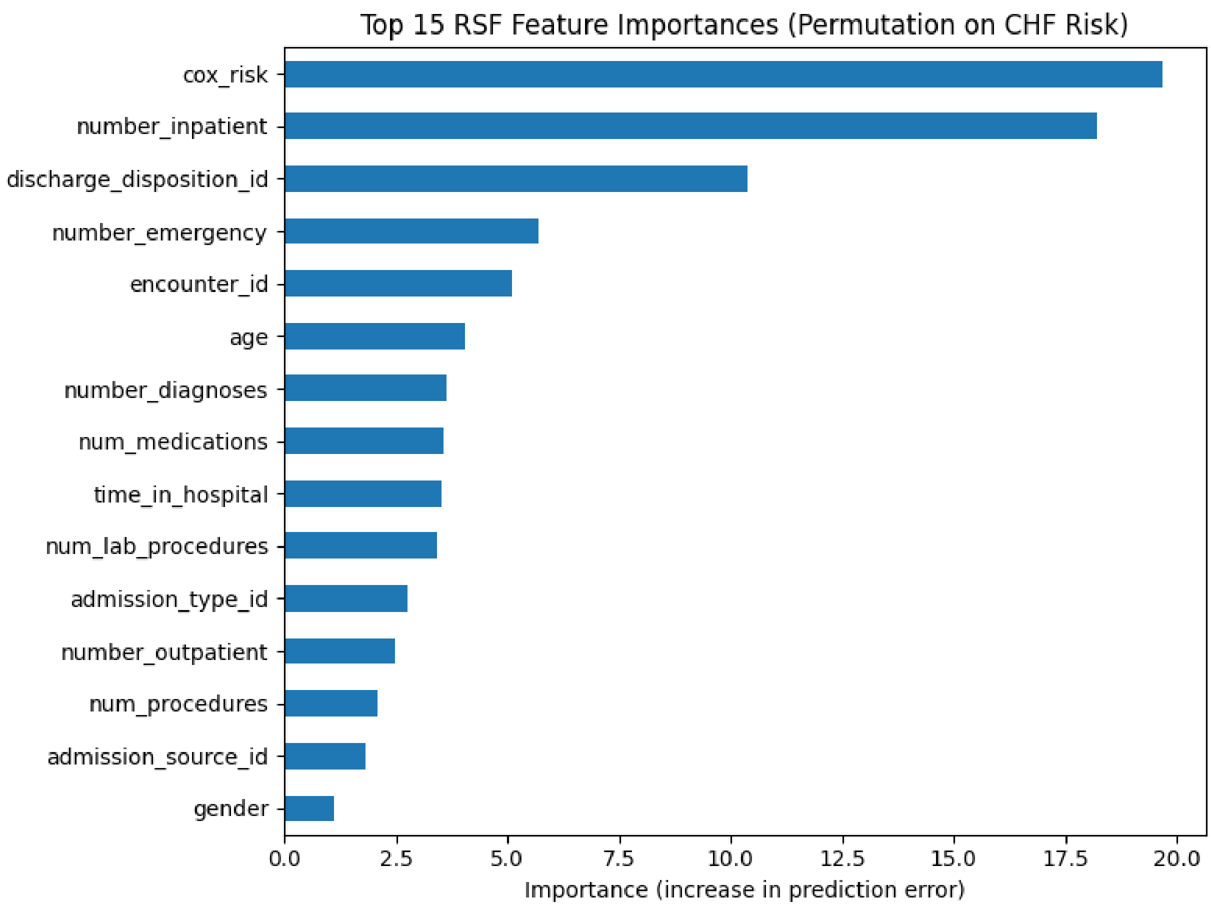

Figure 2. Random Survival Forest Feature Importances (Permutation-Based).

The RSF model identifies **cox_risk**, **number of inpatient visits**, and **discharge disposition** as the strongest predictors of readmission survival time. Higher permutation importance indicates greater increase in prediction error when the variable is shuffled, meaning the feature contributes substantially to accurate risk estimation.

RSF Feature Importance Interpretation

The permutation importance plot identifies the strongest predictors of readmission risk:

1. **cox_risk**

   - The model leverages the Cox-derived risk score as a powerful meta-feature.

2. **number_inpatient**
   - A large increase in the prediction error when permuted indicates strong predictive influence.
   - More prior inpatient visits suggest unstable disease or poor disease control.
3. **discharge_disposition_id**
   - Reflects post-discharge planning quality.
   - Certain dispositions (e.g., SNF, home health) predict higher readmission.
4. **number_emergency**
   - Frequent ED utilization signals acute instability.
5. **age, encounter_id, number_diagnoses**
   - Older patients and those with multiple diagnoses have elevated hazard.

The importance rankings are **consistent with the Cox model**, reinforcing the reliability of the signal.
Clinically, these predictors align with known risk factors for diabetes-related readmission.

3. FULL COMBINED (Executive + Technical + Appendix)

Appendix A — Detailed Model Outputs

Sample Risk Stratification Output

time	event	cox_risk	risk_group
188	0	1.18	High
297	0	1.11	High
112	0	0.71	Low
356	0	0.94	Medium
299	0	0.83	Low

Schoenfeld Residual Test Table

Variables with $p < 0.05$:

- encounter_id
- admission_type_id (borderline)

- gender (borderline)

All other covariates stable ($p > 0.05$).

Appendix B — Interpretability Summary

- **Number of Inpatient Visits** is consistently the strongest predictor → frequent admissions likely reflect uncontrolled disease.
- **Discharge Disposition** strongly influences risk → inadequate outpatient planning.
- **Emergency Visits** mark unstable chronic conditions.
- **Age** shows linear risk increase.

Appendix C — Business Recommendations

1. **High-risk cohort follow-up within 10–14 days** post discharge.
2. **Flag patients with ≥2 prior inpatient visits** for enhanced chronic disease management.
3. **Use RSF model as primary risk stratification tool** in clinical workflow.
4. **Deploy automated alerts** for high-risk discharge dispositions.